Causal Inference for Extreme Events

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Background

- Extreme climate events, such as as droughts, are an escalating global threat, yet they remain difficult to assess due to complex human-biophysical interactions and sector-specific impacts.
- While traditional Machine Learning (ML) yields high predictive accuracy, it often relies on spurious correlations and fails to distinguish true physical drivers; effective climate policy requires a transition toward causal understanding rather than mere association.

Objective: This challenge focuses on implementing **causal inference** methods to estimate the heterogeneous causal effects of meteorological and land-surface factors on the severity of extreme weather events.

Background

The best way to estimate causal effects are through interventional studies that employ randomization (Randomized Controlled Trials, RCTs). RCT studies are not always feasible and we might want to estimate causal effects from observational data.

- Observational study: passively measuring a system of interest without any intervention in the data-generating process
- Interventional study: intentionally influencing data generating process for a particular goal

When using observational data, we must look out for systematic differences between treatment and control groups

Prediction vs Intervention

P (Y | X=x)

Correlation-based
Excels at finding statistical associations

	Predictive Machine Learning	Causal Inference
Goal	**Prediction:** Forecasting future states based on historical patterns.	**Intervention:** Quantifying the effect of targeted actions to change outcomes.
Core Question	What will happen next? (e.g., Will there be a drought?)	**What if we change X?** (e.g., How much crop loss is prevented if we deploy irrigation early?)
ESS Application	**Observational Mapping:** Identifying areas of low soil moisture.	**Impact Mitigation:** Evaluating the true effectiveness of agricultural management responses during extreme droughts.

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P (Y | do(X=x))

Causality-based
Can explain the underlying structures "what if"

What is causal inference (CI)?

CI aims at answering **causal questions** not just statistical ones.

CI techniques provide a means to evaluate treatment effects by controlling for confounding factors, thereby removing associated biases[5]. It isolates direct cause-and-effect relationships to determine how actively intervening on one variable (*treatment/cause*) will change another (*outcome*)[1,2].

The causal effect is the difference Y (T=1) – Y (T=0).

In observational studies we cannot observe both scenarios for the same instance simultaneously, we use statistical methods and assumptions to estimate this difference[5].

The goal is to isolate the effect of the cause as if we had a controlled experiment to estimate average treatment effects (ATE) and conditional average treatment effects (CATE).

What is causal inference (CI)?

The question changes from

What will the land surface temperature be?

to

What would the land surface temperature be if we reforested this area?

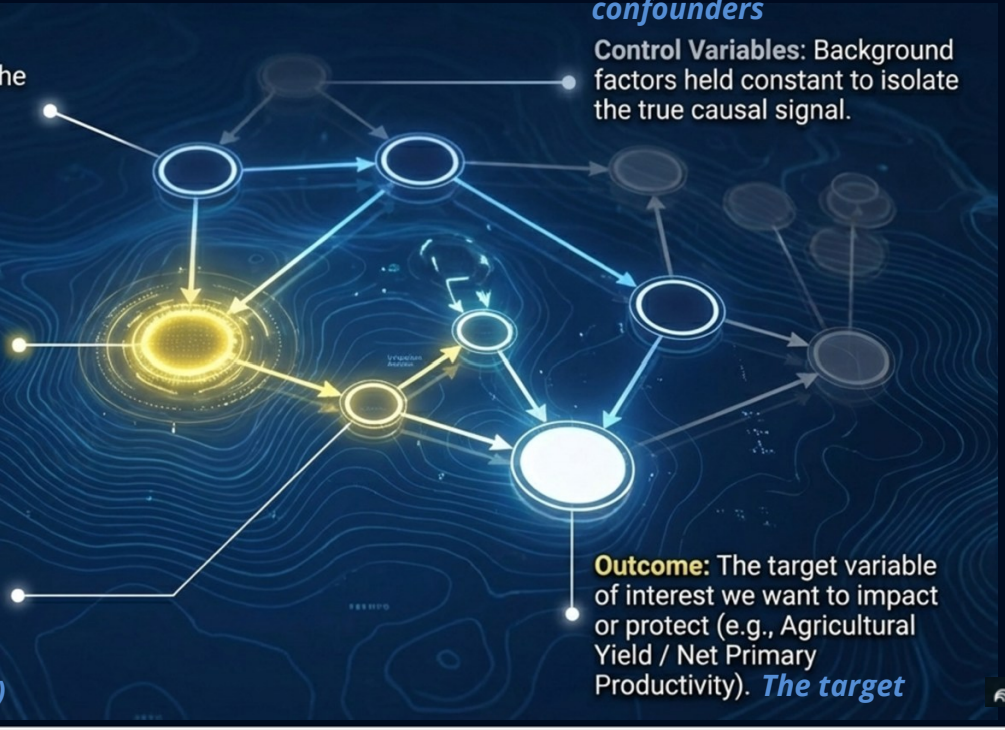
Definitions in a causal relationship

Represent the measured confounders

Control Variables: Background factors held constant to isolate the true causal signal.

Confounders: Variables that influence both the treatment and the outcome (e.g., Soil Type, Altitude, Pre-existing Climate), which must be controlled to block non-causal pathways.

The noise



Treatment: The primary intervention or exposure variable whose effect we want to measure (e.g., Extreme Drought Event).

The cause

Moderators: Variables that affect the strength or direction of the causal relationship (e.g., Access to affordable irrigation modifying the drought's impact).

The context (effect modifiers)

Outcome: The target variable of interest we want to impact or protect (e.g., Agricultural Yield / Net Primary Productivity). *The target*

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Causal inference assumptions

It is possible to estimate causal effects from observational data as long as certain assumptions are met:

- The consistency assumption states that an individual's potential outcome under the observed treatment is equal to the observed outcome
- The ignorability, exchangeability or no-unmeasured confounders assumption states that potential outcomes are conditionally independent of the treatment, given a set of measured covariates. This means that all possible factors (confounders) that are common causes of both T and O are available in the dataset
- The overlap or positivity assumption states that the conditional probability of receiving a treatment (propensity score) should always be greater than 0 and less than 1 for each subgroup defined by various combinations of covariates

Causal effect estimation

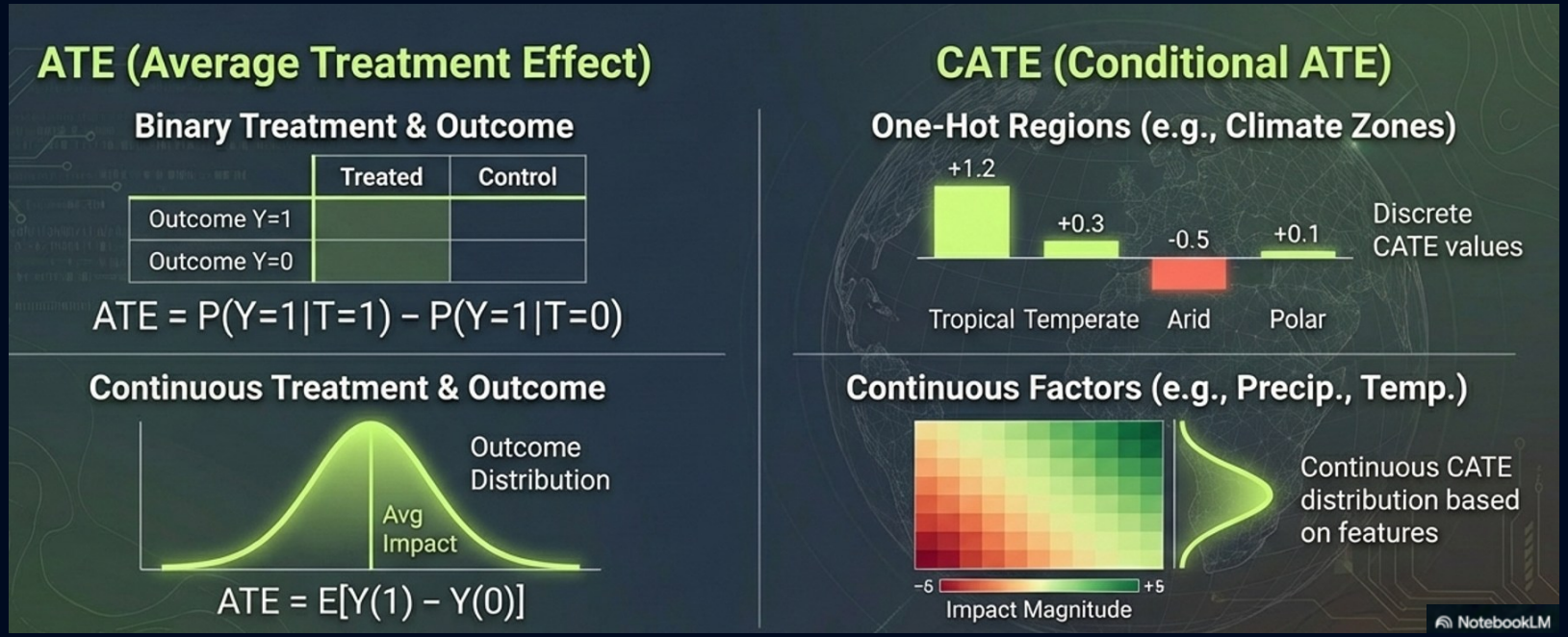
Average Treatment Effect (ATE): the average causal effect of a predictor variable (T) on a response variable of interest (Y)[3]

Conditional average treatment effect (CATE) : captures heterogeneity by estimating the average effect for a specific subgroup defined by particular covariates (X=x)[3]

ATE measures overall impact, and does not explain how different subgroups respond uniquely to an intervention. An intervention might be effective for some subgroups, but it could have no effect in others.

We can measure *heterogeneous* treatment effects, assessing ATE at the subgroup level, which enables more strategic and precise recommendations[5].

Causal effect estimation



Treatment effect based on data type

Treatment

- **Binary**: ATE is the expected difference in the outcome between the treated state (T=1) and the control state (T=0)
- **Continuous**: ATE the average change (marginal effect) in the outcome for each unit increase in the treatment

Outcome

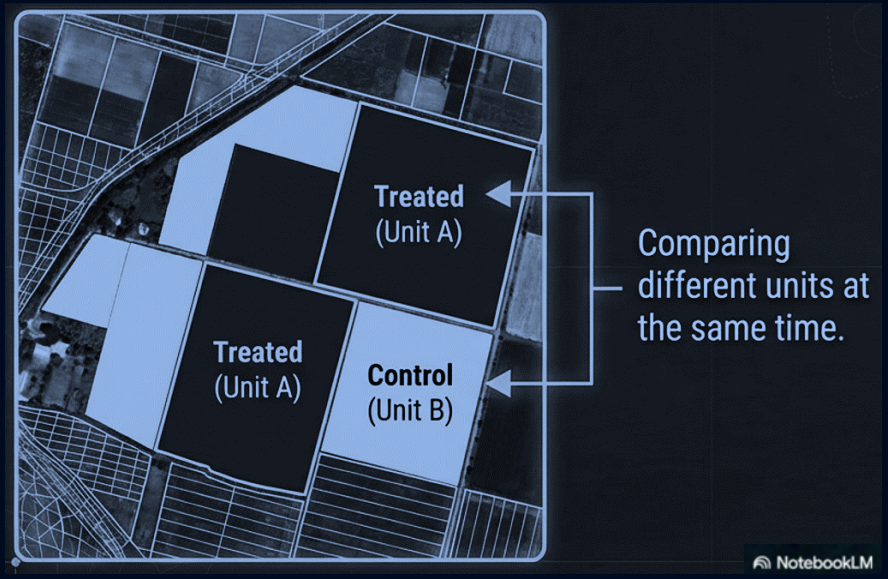
- **Binary**: ATE represent a difference in probabilities, how much the treatment changes the likelihood of the outcome occurring
- **Continuous**: ATE represent the absolute expected change in the outcome variable. CATE forms a continuous distribution across the units

Moderators

- **Categorical** (one-hot encoded): When heterogeneous variables are categorical (such as distinct climate zones), the algorithm groups the units into discrete subgroups. The CATE will be local ATE averages specific to each category
- **Continuous**: the model looks for thresholds that best split the data into subgroups with different causal effects. It averages the trees to generate a continuous spectrum of individual-level CATE estimates for each unit

Cross-sectional analysis

- The “snapshot” approach, examines observational data collected at a single point in time[1], each sample is an independent unit. Time series violates this.
- Longitudinal data, panel data, or time-series methods better account for temporal dynamics
- The challenge: it requires all covariates to simulate a credible control group without longitudinal data



SUTVA assumption: units are independent

Flattening time with lagged features, transform the time-series problem into a static one

Cross-sectional analysis

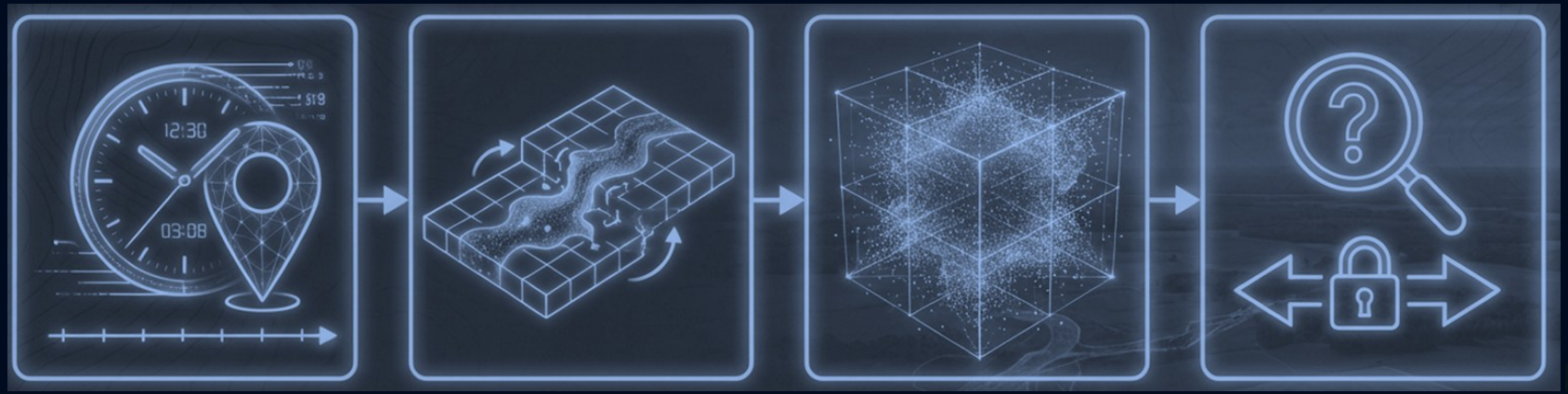
Example:

- Outcome event (Yt_0) – drought status at month t_0
- Treatment (T) and Confounders(W, X) in previous months (t_{-3} to t_{-1})
  - T = soil moisture anomaly in the 3 months leading up to month t (t_{-3} to t_{-1})
  - W_1 = mean temperature anomaly (t_{-3} to t_{-1})
  - W_2 = precipitation anomaly (t_{-3} to t_{-1})
  - W_3 = mean large scale oscillation index (t_{-3} to t_{-1})
  - X = Climate zone

Data preparation: no data leakage from future into past, handling missing data, de-seasoning, scaling, stratify by spatial/temporal blocks.

Challenges in Earth Systems

Spatiotemporal complexity	Spillover (interference)	Dimensionality	Data limitations & Validation
Time lags and location dependence and inherent	Interventions affecting neighboring untreated regions violate independence	High-dimensional covariate spaces that overwhelm and break traditional parametric statistics	Unobserved confounding, selection bias, and difficulty proving robustness and directionality



Challenges in Earth Systems

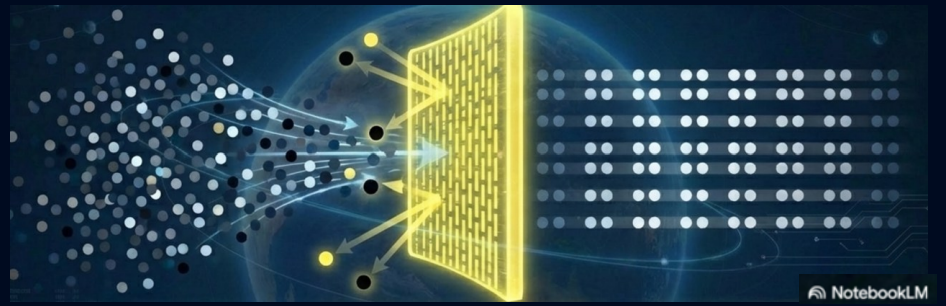
- Standard assumptions often fail in ES due to interconnectedness, time lags, and data limitations.
- Confounding bias is a statistical bias that occurs when a third variable (a confounder) influences both the predictor (treatment) and the outcome variable. If left unaddressed, this shared cause creates a spurious, non-causal statistical association between the treatment and the outcome, making it appear as if the two are directly related[3].

How to tackle it?

- Directed Acyclic Graphs (DAGs) to map out hypothesized causal relationships. Backdoor criterion: identify the minimum set of variables that must be controlled to block non-causal pathways
- Double Machine Learning (DML) eliminates the bias of measured confounders

Methods: Propensity Scores (PS)

- PS is the probability of a unit receiving a treatment (T) given a high-dimensional set of confounding variables -> P (Treatment | Confounders)
- By filtering out extreme PS values (e.g., units with PS < 0.1 and PS >0.9) we enforce the overlap assumption
- We ensure that treated and untreated units being compared are genuinely similar in their characteristics, stripping away selection bias before the ML touches the outcome

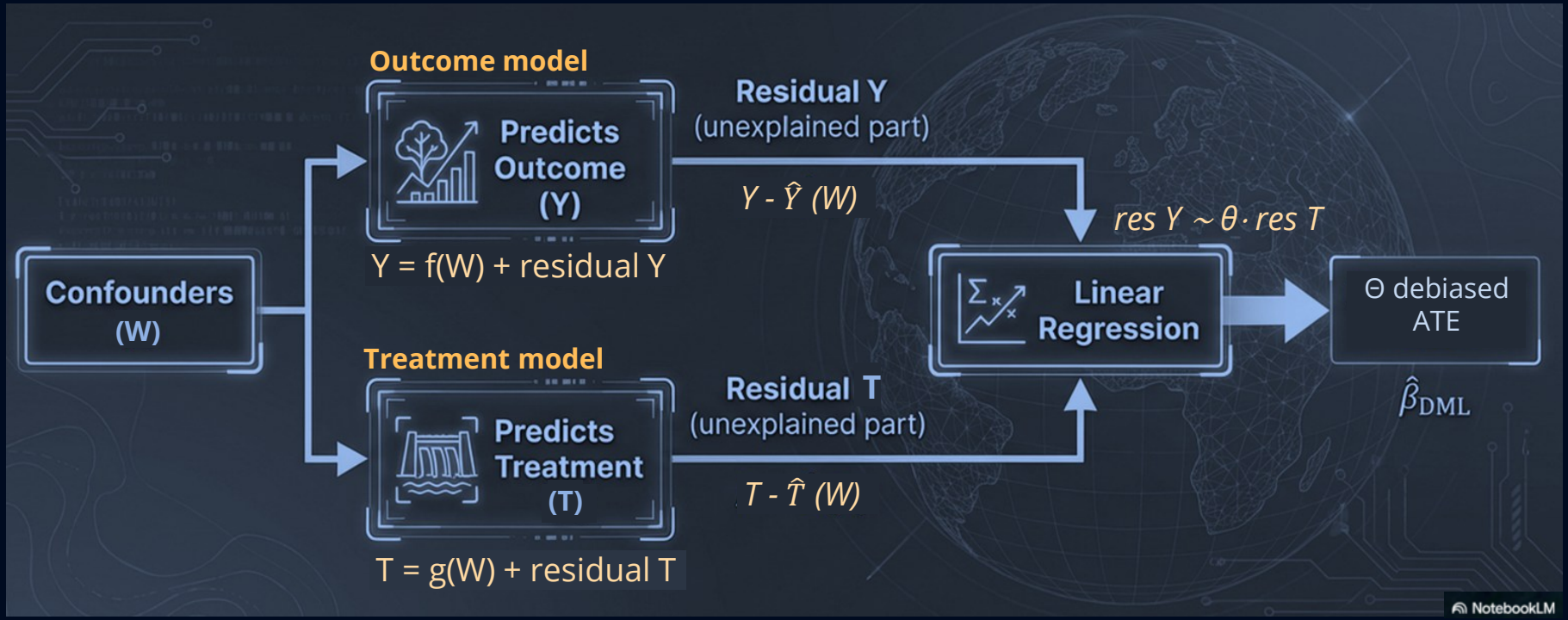


Methods: Double Machine Learning (DML)

DML is a semiparametric causal inference technique that uses ML to estimate ATE using measured confounders by capturing complex, nonlinear relationships.

It achieves this by using a procedure called cross-fitting to separately predict the treatment and the outcome from the covariates, and then isolates the causal effect by regressing the residuals from these two models to remove confounding bias[3]

Methods: Double Machine Learning (DML)



Methods: Double Machine Learning (DML)

Nuisance models are ML models used to separately predict the treatment (T) and the outcome (Y) based solely on the observed covariates (W).

The algorithms can isolate the residual variation in the treatment and outcome that is independent of confounding factors, allowing for an unbiased estimation of the causal effect in the final stage

Outcome model $Y = f(W) + \text{residual } Y$

Treatment model $T = g(W) + \text{residual } T$

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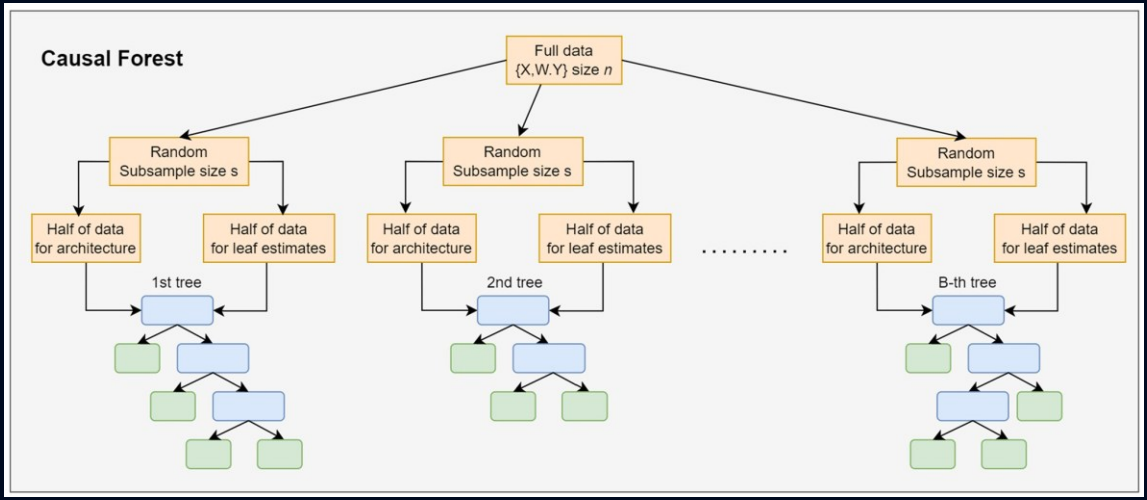
Methods: Causal Forest (CF)

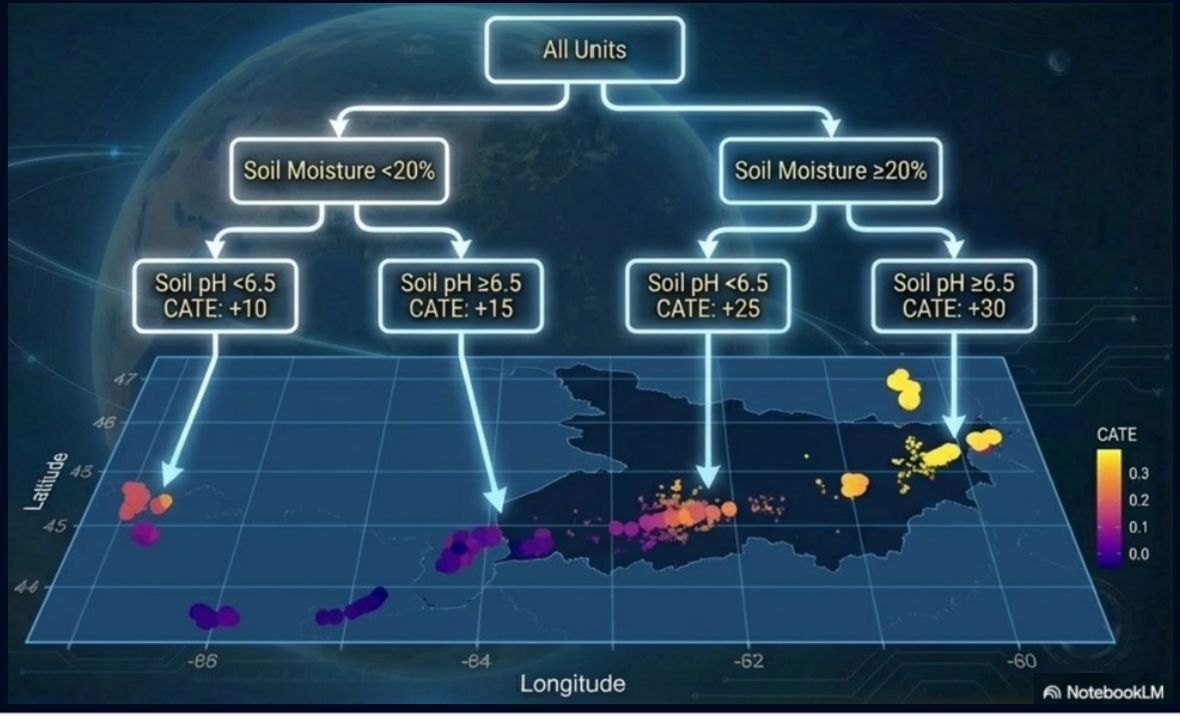
- CF is a random forest modified for causal inference
- Estimates how a factor causes an outcome and captures heterogeneity in effects (CATE)
- CF splits nodes to find the largest differences in the treatment effect based on the moderators, using an “honest” approach, where separated subsets are used to build the tree and evaluate, preventing overfitting



Source: (Duong et al., 2025) [5]

Methods: Causal Forest (CF)

- By averaging over all trees, the CF can estimate a CATE for each set of conditions, telling us how the effect varies across space



Methods: combining DML + CF

- CF require the relationship between the treatment and the outcome to be de-confounded before the tree structure is built
- Confounders are addressed through a process called orthogonalization of residuals
- Researchers often run a DML process first and feed the orthogonalized (de-confounded) residuals directly into the CF
- In CF moderators are introduced as the covariates evaluated at every “decision node” to split the data

Validation / Sensitivity analyses

In causal ML frameworks like DML, validation strategies (often called refutation tests or robustness checks) are used to test the underlying causal assumptions and ensure the estimated effects are reliable[4]

Validation strategies:

- Placebo Treatment: randomly permuting (shuffling) the treatment variable in the dataset. The newly assigned "placebo" treatment has no actual relationship with the outcome, a robust model should return an estimated causal effect that drops to zero.
- Unobserved Common Cause / Sensitivity Analysis / Omitted Variable Test: simulates the presence of an unmeasured confounder (omitted variable) that affects both the treatment and the outcome. Quantify how much the final estimate would change under various strengths of this unobserved confounding, helping determine if the conclusions would hold up even if they missed a variable.

Validation / Sensitivity analyses

Validation strategies:

- Random Common Cause: A random noise variable is added to the dataset as an additional mock confounder. Because the variable is just noise, the estimated causal effect should remain unchanged and stable
- Random Subset Removal: A random subset of the observations is removed from the dataset, and the model is re-run. The estimated effect is expected to remain consistent, showing that the results are not being driven by a specific, small cluster of data points

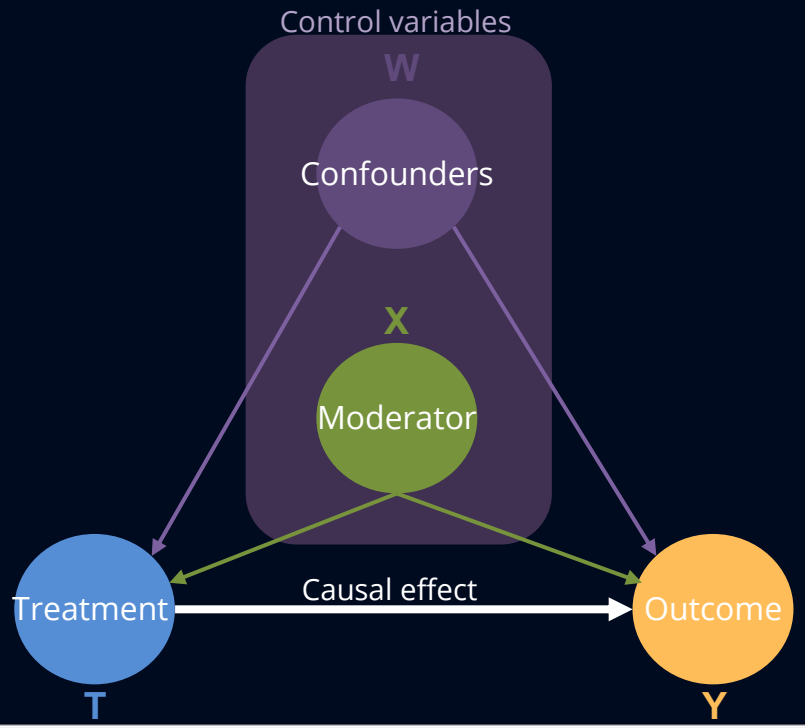
Proposed research questions

How does soil moisture deficits influence the likelihood of agricultural drought impacts, and how do these effects vary across climate types?”

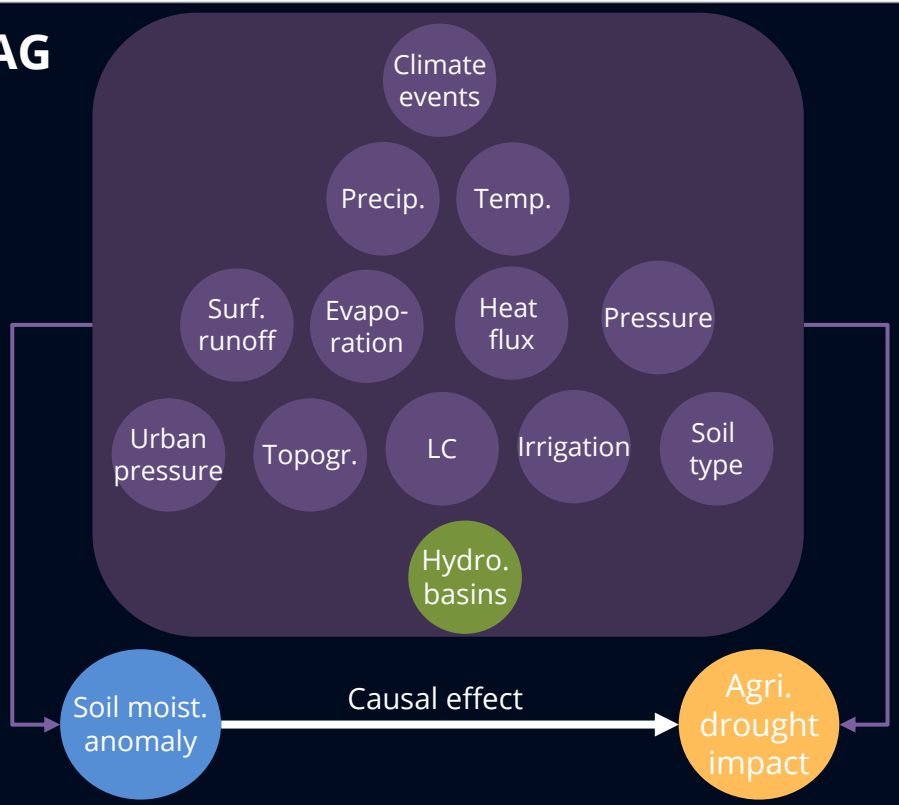
How does meteorological droughts (SPI) influence the likelihood of agricultural drought impacts, and how do these effects vary across hydrological basins?”

How does a "factor" influence the likelihood of "agricultural drought impacts", and how do these effects vary across "regions"?”

An exemplary Directed Acyclic Graph (DAG)



An exemplary DAG



T treatment
Y outcome
W confounders
X moderator

Dataset for the challenge

Multimodal and harmonized dataset. It includes data from several sources.

Data description (link)

- Datacube (zarr) – for visualization
- Tabular data (csv) – for causal effect estimation

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Agricultural drought impact (ADI) – potential outcome

- Monthly ADI with severe and extreme impact (severity) and a duration of at least 60 days (duration) (extreme droughts that had a significant impact on the agricultural sector).
- Events meeting these conditions are aggregated monthly to identify if a month had a reported impact in EDID: European Drought Impact Database.
- If the ADI is negative, but other sector had DI, we discard it to make sure our negatives are negatives.
- To minimize temporal autocorrelation, we use data only for JULY (T) - month with more reported droughts.

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Agricultural drought impact (ADI) – potential outcome

Drought impact in Agriculture sector (EDID)

- Annual Crops: Drought reduces soil moisture and disrupts the growth cycle, leading to lower yields, poor germination, and greater vulnerability to pests and diseases. This forces farmers to use irrigation, drought-tolerant varieties, and soil conservation practices.
- Permanent Crops: Persistent drought stresses perennial plants by reducing water availability, cutting fruit set, and increasing the risk of tree mortality. Long-term effects include diminished productivity and gradual decline in plant health, prompting enhanced irrigation and maintenance strategies.
- Livestock: Water shortages and reduced feed availability under drought conditions result in heat stress, decreased reproduction, and higher disease risk among animals. Producers often respond by employing water conservation, nutritional adjustments, and choosing more drought-resilient livestock breeds.

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Agricultural drought impact (ADI) – potential outcome

edid EUROPEAN DROUGHT IMPACT DATABASE

EXPLORE SEARCH ABOUT

EDID - Search drought impact records

Drought impact record 39184

Sector
Agriculture - Permanent crops

Severity
Extreme

Time span
From 2016-10-01 until 2016-12-31

Description
However, and in global terms, it is the rainfed productions that are taking the worst part. Thus, in the olive grove there is a decrease in tree yields of 45%, a drop in production of 72,000 tons and a decrease in income of 38.3 million euros

Reference
20 Minutos, https://www.20minutos.es/noticia/2861015/0/ava-asaja-cifra-245-millones-perdidas-por-sequia-sector-agropecuario-exige-medidas-excepcionales/

Sources
EDORA

Record created on 2023-11-03, last updated on 2023-11-03

Location

Affected countries
Spain

Affected NUTS1
Este

Affected NUTS2
Comunitat Valenciana

Affected NUTS3
Alicante/Alacant, Castellón/Castelló, Valencia/València

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Agricultural drought impact (ADI) – potential outcome

Most Common impacts

- Significant yield reductions and crop damage: Corn (maize), cereals, wheat, potatoes, sunflowers, vegetables, fruit trees, olives, and viticulture. Harvest losses often ranged from 25% to 60%
- Livestock and food for livestock shortages: A critical lack of fodder (hay, grass, and corn silage) was reported. There were notable decreases in milk production and quality. Farmers were frequently forced to send cattle to slaughterhouses earlier.
- Economic and financial losses: Damages often totaled hundreds of millions of euros. The scarcity of feed led to "growing prices" for fodder, which in turn increased the cost of milk and meat. Numerous reports mention farmers requesting EU or state compensation and the declaration of "natural disasters."
- Operational impacts: Earlier-than-usual harvesting of winter cereals and grapes (to save what was left). Water restrictions and the drying up of wells, reservoirs, and rivers used for irrigation.

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Agricultural drought impact (ADI) – potential outcome

Reasons for the impacts

The descriptions consistently point to a combination of meteorological factors:

- Precipitation deficit: The primary driver was a lack of rain during critical periods (winter/spring 2011/2012, summer 2015, etc.).
- High temperatures and heatwaves: Extreme heat (often exceeding 40°C) accelerated evaporation and caused direct “heat stress” to crops and animals.
- Poor timing: Droughts often followed a dry winter, leaving no soil moisture reserve for the growing season. In some cases, a wet start to the season followed by an extremely dry July created a “critical shortage” by the end of the summer.
- Dry spells: Prolonged periods without rain (mid-July to mid-August) were frequently mentioned as the cause for “irreversible damage” to summer crops.

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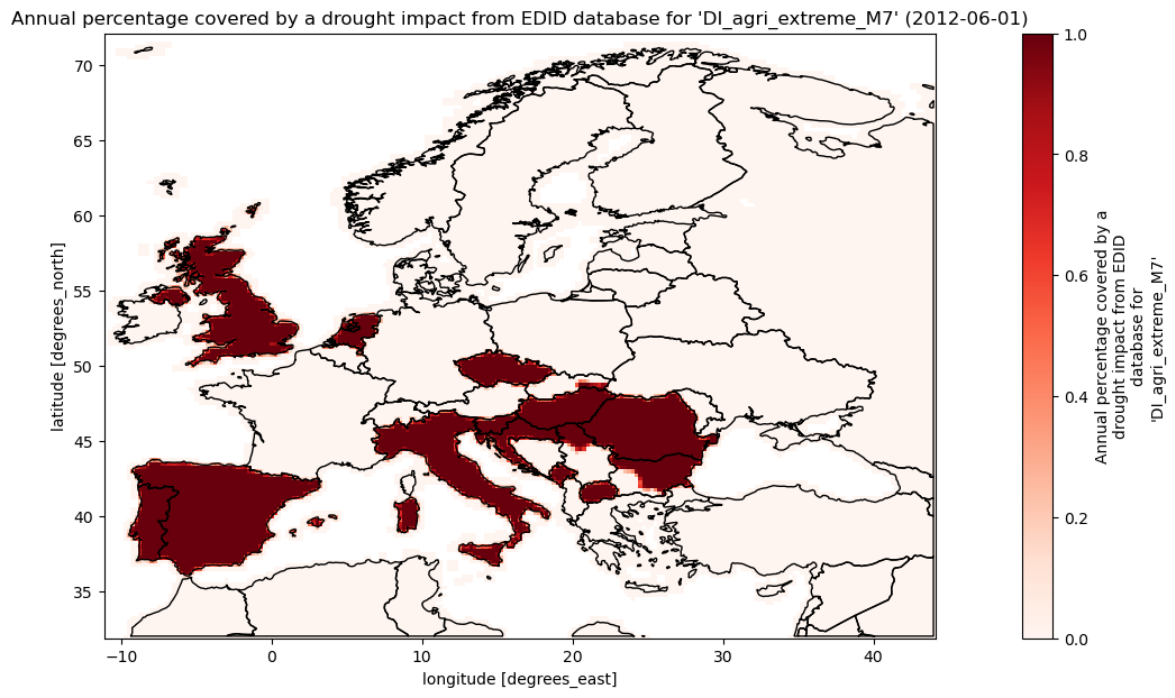
Agricultural drought impact (ADI) – potential outcome



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Meteorological and land-surface factors

- Monthly aggregated values for +60 climate and environmental drivers
- Resample all variables to “ERA5” 0.25 degrees
- Remove seasonality and keep anomalies (removing monthly means)
- Standardization (z-scores)
- To minimize temporal autocorrelation, we use data only for JUNE (T-1), the previous month of the outcome, to avoid data leakage from future into past.
- To minimize spatial autocorrelation, we do a regional sampling

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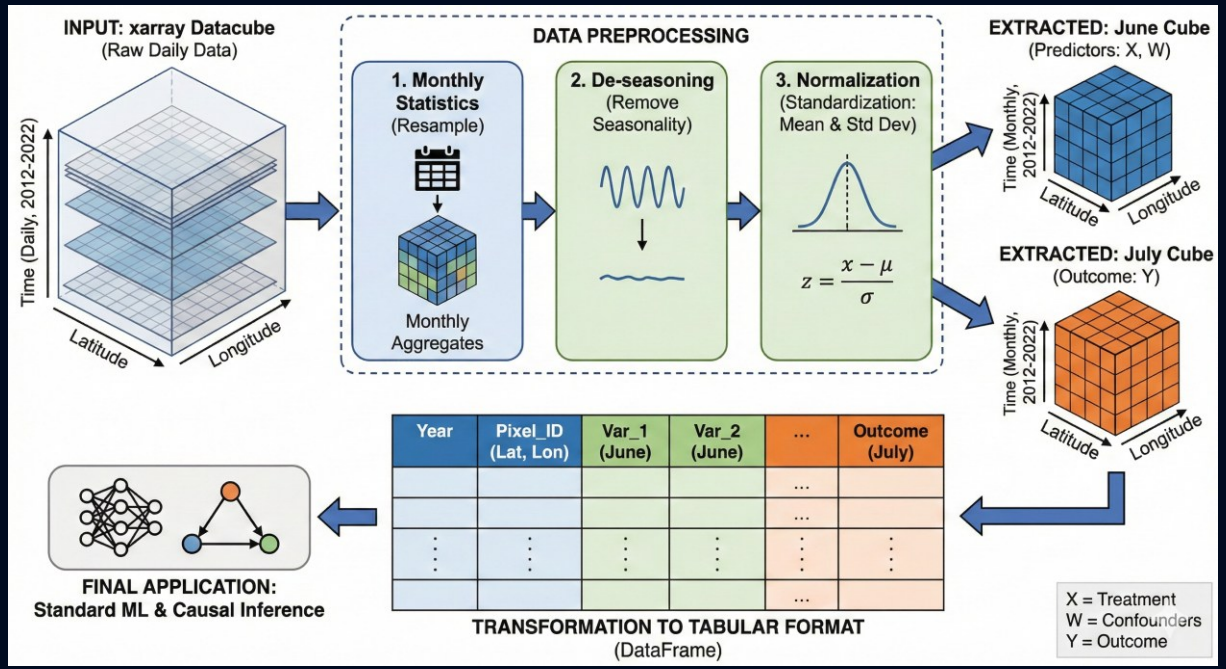
Preprocessing



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Preprocessing - sampling

- We propose a regional sampling, instead of a random sampling.
- We enforce a regional sampling strategy with a minimum distance constraint.
- Ensuring a minimum distance between samples, guarantees that points are well-distributed while maintaining randomness.
- Our data has a pixel size of 0.25 degrees. We use 0.50 (2x2)
- In each timestep we move the location of the pixel, to avoid having the same pixel over time.

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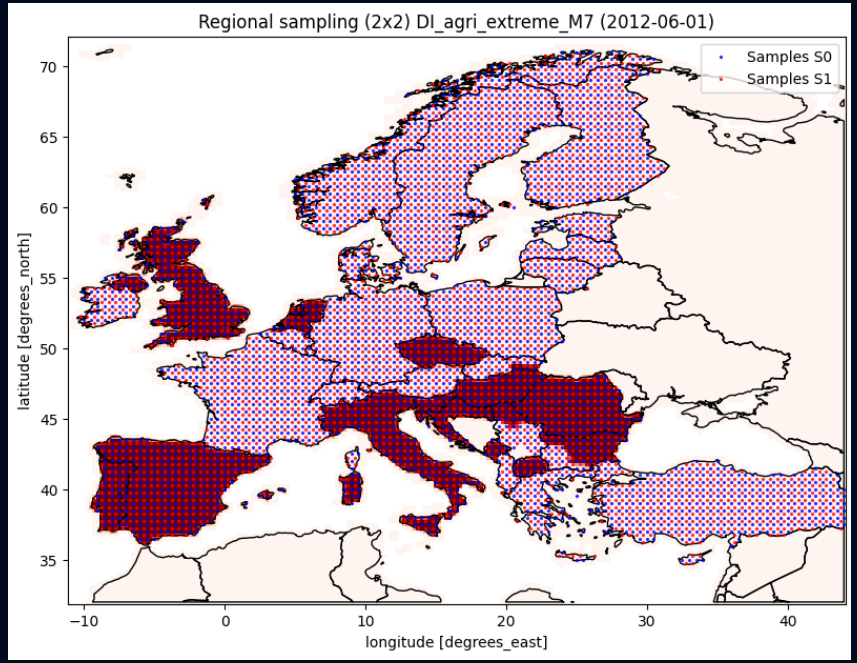
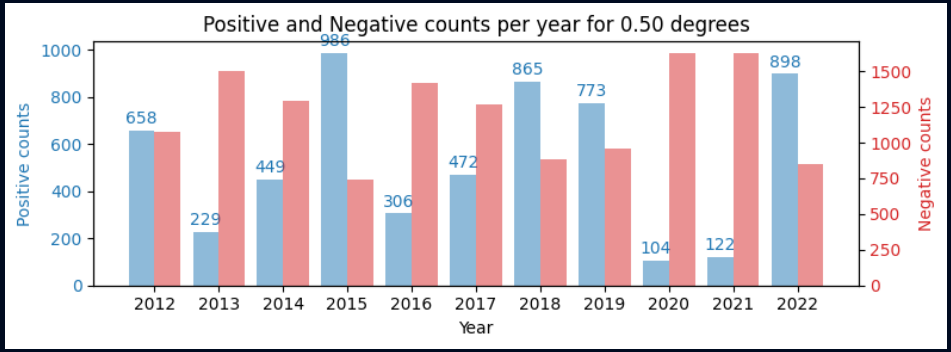
Preprocessing - sampling





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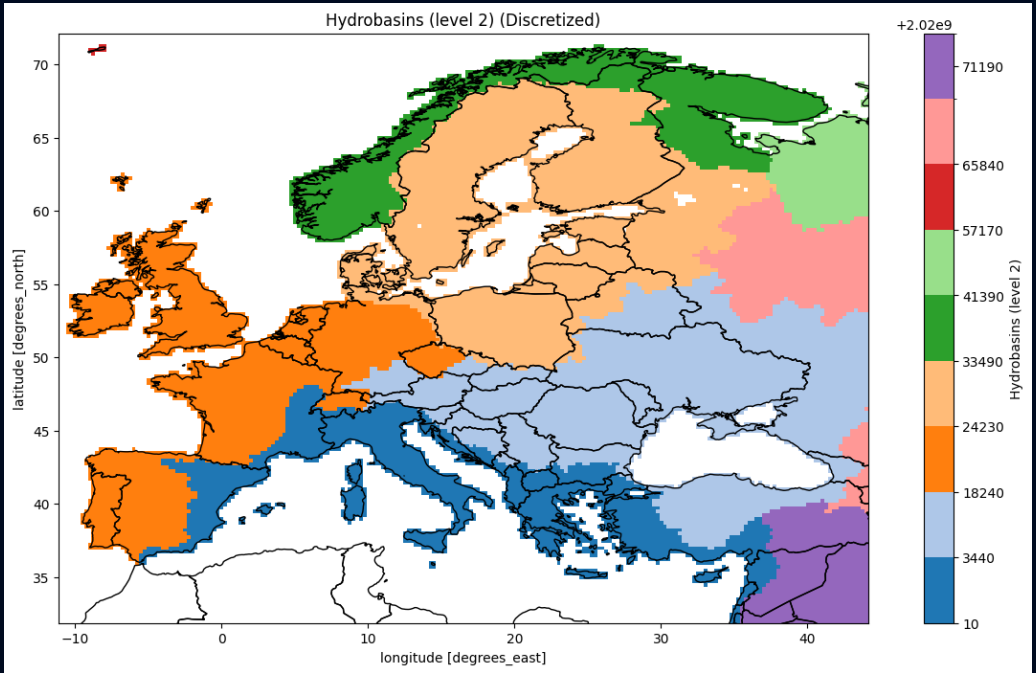
Heterogenous effect, example



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